

ADAPTIVE INTELLIGENT TRANSPORTATION SYSTEM

Project Overview

The Adaptive Intelligent Transportation System is an IoT-based smart traffic management solution developed to improve emergency vehicle movement in urban traffic conditions. The system focuses on creating a smart ambulance corridor using intelligent divider control mechanisms and automated traffic signal management.

The proposed system detects emergency ambulances using sensors and automatically activates divider movement and traffic prioritization mechanisms to provide a clear route for emergency transportation. The system also integrates GSM and GPS technologies for real-time alert communication and location tracking.

This project aims to reduce ambulance delay during emergencies and improve response time in densely populated traffic environments.

Objectives

- To design an intelligent ambulance priority system
- To automate divider control using ESP32
- To reduce ambulance waiting time in traffic congestion
- To provide emergency alert notifications using GSM
- To integrate GPS-based location tracking
- To improve smart city transportation infrastructure

Technologies Used

- ESP32 NodeMCU
- Embedded Systems
- IoT Technology
- GSM Communication
- GPS Tracking
- IR Sensors
- Servo Motor Automation
- LCD Display Module
- Arduino IDE

Hardware Components

Component	Purpose+
ESP32 NodeMCU	Main Controller
IR Sensors	Vehicle Detection
Servo Motor	Divider Control
GSM Module	SMS Alerts
GPS Module	Location Tracking
16x2 LCD	Status Display
LEDs	Traffic Signal Indication
Power Supply	System Operation

System Methodology

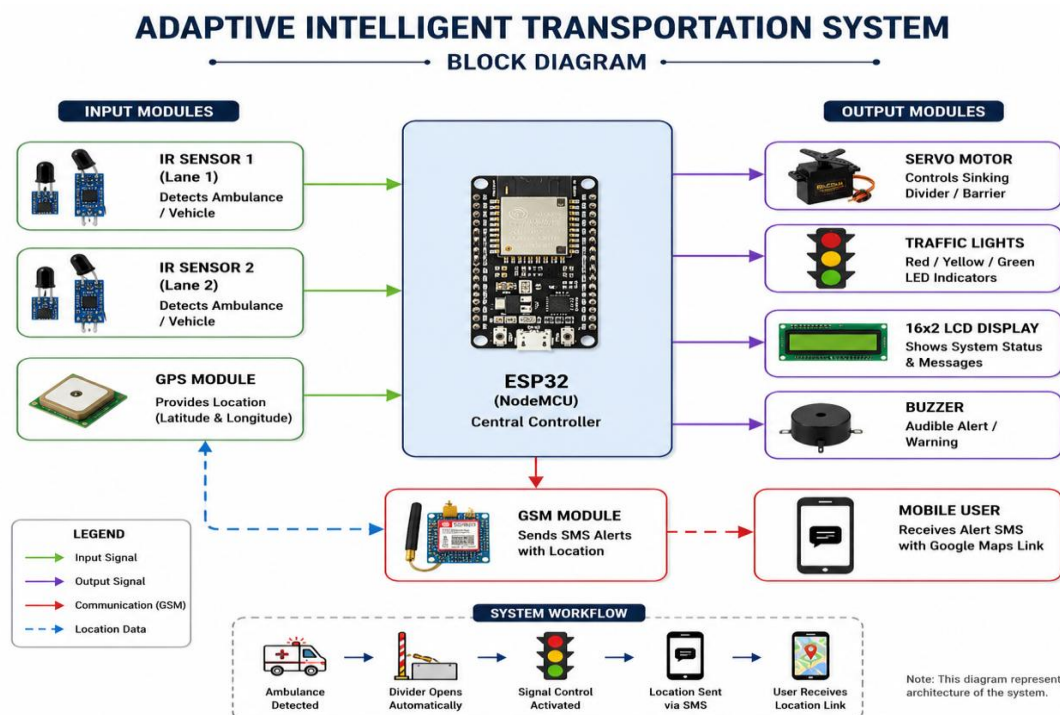
The system continuously monitors vehicle movement using IR sensors placed near the traffic divider region. When an ambulance is detected, the ESP32 controller activates the intelligent control mechanism.

The servo motor operates the divider system to create an emergency path for the ambulance. Simultaneously, traffic signal indicators are adjusted to prioritize ambulance movement.

The GSM module sends emergency notifications to predefined mobile users, while the GPS module provides live location coordinates for monitoring and navigation purposes.

The LCD module displays system status messages such as ambulance detection, waiting notifications, and normal traffic mode.

Block Diagram



Features

- Intelligent ambulance detection
- Automatic divider control
- Real-time traffic prioritization
- Emergency SMS alerts
- GPS-based location sharing
- Smart traffic signal management
- LCD status monitoring
- IoT-enabled automation

Applications

- Smart City Infrastructure
- Emergency Medical Services
- Intelligent Traffic Systems
- Urban Transportation Management
- Highway Emergency Corridors
- Public Safety Systems

Advantages

- Reduces ambulance delay
- Improves emergency response time
- Enhances traffic efficiency
- Supports smart transportation
- Low-cost IoT implementation
- Real-time monitoring capability

Future Enhancements

- AI-based vehicle recognition
- Cloud server integration
- Mobile application dashboard
- Real-time traffic analytics
- Multiple ambulance coordination
- Smart city network integration

Code Availability

Only a simplified public demonstration version of the implementation is available in this repository.

The complete implementation details and source code are intentionally withheld due to ongoing research publication and intellectual property considerations.

Adaptive Intelligent Transportation System - Sample ESP32 Logic

```
#include <ESP32Servo.h>
```

```
Servo divider;
```

```
int irSensor = 32;
```

```
int servoPin = 4;
```

```
void setup() {
```

```
  pinMode(irSensor, INPUT);
```

```
  divider.attach(servoPin);
```

```
  divider.write(0);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
  int vehicleDetected = digitalRead(irSensor);
```

```
if(vehicleDetected == LOW) {  
  
    ...  
  
    Serial.println("Emergency vehicle detected");  
  
    divider.write(90);  
  
    delay(3000);  
  
    divider.write(0);  
    ...  
  
}  
}
```

Conclusion

The Adaptive Intelligent Transportation System demonstrates an effective IoT-based approach for improving ambulance movement in traffic-congested areas. The system combines embedded systems, automation, and communication technologies to create a smart emergency corridor solution.

The project highlights the potential of intelligent transportation systems in enhancing urban emergency management and smart city infrastructure development.

Repository Information

GitHub Repository: <https://github.com/Suriya2112/adaptive-intelligent-transportation-system.git>

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Domain: Internet of Things (IoT) and Smart Transportation Systems